

$$\begin{bmatrix} 1 & -2 & 0 & 0 & 0 \\ 0 & -3 & 1 & 0 & 5 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -3 \end{bmatrix} \xrightarrow[\text{-Row 3}]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 1 & -2 & 0 & 0 & 0 \\ 0 & -3 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -3 \end{bmatrix} \xrightarrow[\times (-1/3)]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 1 & -2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -3 \end{bmatrix}$$

$$\xrightarrow[\text{+2} \times \text{Row 2}]{\text{Row 1} = \text{Row 1}} \begin{bmatrix} 1 & 0 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -3 \end{bmatrix} \quad \therefore x = \begin{bmatrix} -2 \\ -1 \\ 2 \\ -3 \end{bmatrix} \quad \square$$

Find an LU factorization of the matrices in Exercises 7-16 (with L unit lower triangular). Note that MATLAB will usually produce a permuted LU factorization because it uses partial pivoting for numerical accuracy.

9. $\begin{bmatrix} 3 & -1 & 2 \\ -3 & -2 & 10 \\ 9 & -5 & 6 \end{bmatrix}$ Solution:

$$\begin{bmatrix} 3 & -1 & 2 \\ -3 & -2 & 10 \\ 9 & -5 & 6 \end{bmatrix} \xrightarrow[\text{+Row 1}]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 3 & -1 & 2 \\ 0 & -3 & 12 \\ 9 & -5 & 6 \end{bmatrix}$$

$$\xrightarrow[\text{-3} \times \text{Row 1}]{\text{Row 3} = \text{Row 3}} \begin{bmatrix} 3 & -1 & 2 \\ 0 & -3 & 12 \\ 0 & -2 & 0 \end{bmatrix} \xrightarrow[\times 1/(-3)]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 3 & -1 & 2 \\ 0 & 1 & -4 \\ 0 & -2 & 0 \end{bmatrix} \xrightarrow[\text{+2} \times \text{Row 2}]{\text{Row 3} = \text{Row 3}} \begin{bmatrix} 3 & -1 & 2 \\ 0 & 1 & -4 \\ 0 & 0 & -8 \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ -3 \\ 9 \end{bmatrix} \div 3 \quad \begin{bmatrix} -1 \\ -2 \\ -5 \end{bmatrix} \div -3 \quad \begin{bmatrix} 2 \\ 12 \\ 0 \end{bmatrix} \div -8$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 3 & 2/3 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 3 & -1 & 2 \\ 0 & 1 & -4 \\ 0 & 0 & -8 \end{bmatrix}$$

11. $\begin{bmatrix} 3 & -6 & 3 \\ 6 & -7 & 2 \\ -1 & 7 & 0 \end{bmatrix}$ Solution:

$$\begin{bmatrix} 3 & -6 & 3 \\ 6 & -7 & 2 \\ -1 & 7 & 0 \end{bmatrix} \xrightarrow[\text{+Row 1}]{\text{Row 3} = \text{Row 3}} \begin{bmatrix} 3 & -6 & 3 \\ 6 & -7 & 2 \\ 0 & 5 & -4 \end{bmatrix} \xrightarrow[\text{-Row 2}]{\text{Row 3} = \text{Row 3}} \begin{bmatrix} 3 & -6 & 3 \\ 6 & -7 & 2 \\ 0 & 5 & 1 \end{bmatrix}$$

$$\xrightarrow[\times 1/3]{\text{Row 1} = \text{Row 1}} \begin{bmatrix} 1 & -2 & 1 \\ 6 & -7 & 2 \\ -1 & 7 & 0 \end{bmatrix} \xrightarrow[\text{-6} \times \text{Row 1}]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 1 & -2 & 1 \\ 0 & 5 & -4 \\ -1 & 7 & 0 \end{bmatrix} \xrightarrow[\times 3]{\text{Row 1} = \text{Row 1}} \begin{bmatrix} 3 & -6 & 3 \\ 0 & 5 & -4 \\ 0 & 0 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ 6 \\ -1 \end{bmatrix} \div 3 \quad \begin{bmatrix} -6 \\ -7 \\ 7 \end{bmatrix} \div 5 \quad \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} \div 5$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1/3 & 1 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 3 & -6 & 3 \\ 0 & 5 & -4 \\ 0 & 0 & 5 \end{bmatrix}$$